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Unit – 5 : Session – 12 : SLO – 2 | LDM/STM, Programmed I/O and Encoding

■ I. Multiple Choice

1. B) LDM — Load Multiple registers.
2. C) Polling — programmed I/O repeatedly checks the device status register.
3. D) DB (Decrement Before) — STMFD = Store Multiple, Full Descending stack = decrement-before addressing.

■ II. Fill in the Blanks

1. status — program-controlled I/O repeatedly checks the device status register.
2. stack — LDMFD restores registers from the (full descending) stack.
3. operand — binary encoding identifies the opcode and operand fields.

■ III. True or False

1. TRUE — polling burns CPU cycles continuously checking the device.
2. FALSE — LDMIA loads in increasing address order (Increment After).
3. TRUE — STR R1, [R2] stores R1's value at the address in R2.

■ IV. Match the Following

- A) STMFD — 2) Store multiple registers.
- B) LDMFD — 3) Load multiple registers.
- C) Polling — 1) Waiting for device ready.
- D) STR — 4) Write to memory.

■ V. Numerical Answers

1. MOV R1, #5 → cond=1110, I=1, opcode=1101 (MOV), S=0, Rn=0000, Rd=0001, immediate=0x005 → 0xE3A01005.
2. 20 bytes — 5 registers × 4 bytes.
3. Polling loop:
 - LDR R2, =0x60000020
 - loop: LDR R0, [R2]
 - TST R0, #1 @ test bit 0
 - BEQ loop @ keep waiting until bit 0 = 1